FLIGHT RIGHT · FINAL YEAR PROJECT · PROGRESS REPORT

Predicting Where Aircraft Will Go

What we built, what comes out of it, how well it works, and why we have changed the model from the one described in our proposal.

WHO THIS IS WRITTEN FOR

This report assumes **no background in machine learning and no background in aviation**. Every technical word is explained the first time it appears. It is written to be read straight through by a mentor or a teammate who has not been involved in the model work.

THE ONE THING TO KNOW BEFORE READING

In our proposal we said we would use a particular type of AI model called a **Bi-LSTM**. We built it. It performed worse than simple arithmetic. We found the reason, corrected it, and the corrected model now works well. **This report explains that change and gives the measurements behind it.** Section 11 sets out exactly what has changed relative to the proposal.

Prepared
4 September 2026

Component
Model 1 of 3 — Trajectory Prediction

Data
18 collection sessions, 24 Aug – 1 Sep 2026

Status
Working; results measured and reproducible

SECTION 1

What the system is trying to do

Air traffic controllers keep aircraft apart. To do that they need to know not only where each aircraft is now, but where it is going. Our project builds a tool that helps with the second part.

Every ten seconds or so, we receive a position report from every aircraft flying near Mumbai. Each report says where the aircraft is, how high, how fast, and which direction it is pointing.

Our system takes the last ten of those reports for a given aircraft, and answers one question:

> **THE QUESTION THE MODEL ANSWERS**
>
> **Where will this aircraft be in 1 minute, 3 minutes, 5 minutes and 10 minutes from now?**

If we can answer that for every aircraft, then we can compare pairs of aircraft and see whether any two are heading for the same piece of sky. If they are, we warn the controller early enough for them to do something about it.

This report covers only that prediction step. The two later parts of the project — deciding whether a warning is needed, and suggesting what to do about it — are separate and are not covered here.

Words used in this report

Only these eight are needed. Everything else is ordinary English.

Nautical mile (NM) The distance unit used in aviation. One nautical mile is about 1.15 ordinary miles, or 1.85 kilometres.	**Separation** The gap controllers must keep between two aircraft. Normally 5 nautical miles apart sideways, or 1,000 feet different in height.
Within 5 NM Our main score. "93% within 5 NM" means: for 93 aircraft out of every 100, our prediction landed less than 5 nautical miles from where the aircraft actually went.	**The simple method** Our comparison yardstick. It assumes the aircraft keeps flying in a dead straight line at its current speed. No AI, just multiplication.
Model The trained AI program. It has been shown many thousands of real flights and has learned patterns from them.	**Training / testing** We let the model learn from flights recorded on earlier days, then measure it on flights from later days it has never seen. This stops it from simply memorising.
Terminal area The busy airspace within roughly 40 to 60 nautical miles of the airport, where aircraft are climbing, descending, turning and queuing to land.	**VABB** The official international code for Mumbai airport. It appears in some of our data tables.

SECTION 2

Where our data comes from, and what is in it

We collect live aircraft position reports from the **OpenSky Network**, a free public service run by a research consortium. Aircraft continuously broadcast their own position by radio, and OpenSky collects those broadcasts from receivers around the world and makes them available.

We ran a collection program that asked OpenSky, every few seconds, for every aircraft inside a box covering the Mumbai region. Each answer was saved as one row in a data file — much like a row in a spreadsheet.

> WHAT WE COLLECTED
>
> **18 collection sessions** of about 90 minutes each, between 24 August and 1 September 2026, producing **329,647 usable position reports** from aircraft in flight.

The columns in each row of our data files

Each row is one aircraft at one moment in time. There are 19 columns. We use eight of them; the rest are recorded but not fed to the model.

EVERY COLUMN IN THE COLLECTED DATA

COLUMN NAME	WHAT IT MEANS IN PLAIN WORDS	USED?
icao24	A permanent code unique to that individual aircraft, like a number plate. We use it to group reports into one aircraft's journey.	Yes
time_position	The exact second at which the aircraft was at this position.	Yes
latitude	How far north or south the aircraft is.	Yes
longitude	How far east or west the aircraft is.	Yes
baro_altitude	How high the aircraft is, in metres.	Yes
velocity	How fast it is travelling over the ground, in metres per second.	Yes
true_track	Which compass direction it is travelling in, in degrees.	Yes
vertical_rate	How fast it is climbing or descending. Negative means descending.	Yes
on_ground	Whether the aircraft is still on the runway or taxiway.	Filter only
callsign	The flight number, e.g. AIC1RH. Not used by the model.	No
origin_country	Country the aircraft is registered in.	No
last_contact	Last moment any signal was heard from the aircraft.	No
geo_altitude	A second, differently-measured height figure.	No
squawk, spi	Radar identification codes set by the pilot.	No
sensors, position_source	Which ground receiver picked up the signal.	No
category	Aircraft size class. Almost always blank in our data.	No
fetch_timestamp_utc	When our own program asked for the data.	No

We discard rows where the aircraft is on the ground, below 1,000 feet, slower than about 100 knots, or where any of the eight needed values is missing. What is left is aircraft genuinely in flight.

> **AN IMPORTANT LIMITATION, STATED UP FRONT**
>
> This data tells us **where every aircraft is**. It does **not** tell us where any aircraft intends to go. We do not receive flight plans, and we do not receive the instructions controllers give by radio. Section 10 explains why this matters, and it is the single biggest limit on what our system can achieve.

SECTION 3

How raw data becomes training examples

A model cannot learn from a pile of position reports. It needs matched questions and answers. We build them like this:

```
Step 1  Sort every position report by aircraft, then by time.
Step 2  Take any ten reports in a row from one aircraft. This is the question: about 100
seconds of recent flying.
Step 3  Look forward in that same aircraft's data to find where it actually was 1, 3, 5 and
10 minutes later. These are the answers.
Step 4  Store the question and its answers as one training example. Slide forward one report
and repeat.
```

This produced **198,537 examples**. We divide them by date, so the model learns on earlier days and is tested on later days it has never seen:

HOW THE EXAMPLES ARE DIVIDED

GROUP	EXAMPLES	PURPOSE
Training	143,117	The model learns from these.
Validation	27,552	Used during training to decide when to stop, so it does not over-memorise.
Testing	27,868	Never seen during training. Every result in this report comes from these.

Two corrections we had to make here

First — the answers were slightly wrong. Position reports do not arrive at exactly 1, 3, 5 and 10 minutes. Originally we took whichever report was nearest, accepting a mismatch of up to 30 seconds. But an airliner covers about 13 kilometres in a minute, so a 30-second mismatch put up to 3.7 nautical miles of error into the answer itself — larger than the error we were trying to teach the model to achieve. We now calculate the exact position between the two nearest reports.

Second — we were throwing away all the landing traffic. Originally an example was kept only if all four answers existed. An aircraft descending into Mumbai drops below our height filter within ten minutes, so its 10-minute answer is missing — and the entire example was deleted. This silently removed almost all arriving aircraft from our training data, which is exactly the traffic that is hardest to predict. We now keep examples with however many answers exist and simply mark which ones are missing.

> **EFFECT OF THE SECOND CORRECTION**
>
> Training examples went from **45,377 to 143,117** — more than three times as many — and the additions were mainly arriving aircraft that had been invisible to the model before.

SECTION 4

What goes into the model — the 20 inputs

The model does not receive the raw columns from Section 2 directly. We convert them into 20 pieces of information that describe *how the aircraft is moving*, not just where it happens to be on a map.

This is described relative to the aircraft itself. "Ahead" means ahead along its current heading; "right" means to its right. This matters: it means a right turn over Pune looks identical to the model as a right turn over Surat, so a pattern learned in one place is usable everywhere.

Part A — the recent movie: 7 measurements at each of the 10 time steps

MEASUREMENT	WHAT IT TELLS THE MODEL
Distance ahead	How far in front of its current position the aircraft was at that moment, in metres.
Distance to the side	How far to the left or right of its current heading it was. This is what reveals a turn.
Height difference	How much higher or lower it was than it is now.
Speed	How fast it was going at that moment.
Climb / descent rate	How fast it was going up or down at that moment.
Heading difference	How much its compass direction has changed since that moment. Positive one way, negative the other, so left and right turns are distinguishable.
Time gap	How many seconds passed since the previous report, because they do not arrive perfectly evenly.

Part B — the summary: 13 numbers describing the whole 100 seconds

SUMMARY NUMBER	WHAT IT TELLS THE MODEL
Current height	High usually means cruising; low usually means arriving or departing.
Current speed	Fast means cruising; slow means approaching.
Current climb / descent rate	Whether it is going up, down, or holding level right now.
Turn rate	How quickly it is turning at this moment — the single strongest warning sign of an unpredictable aircraft.
Speeding up or slowing down	A change of speed usually signals a change of flight phase.
Change in climb rate	Distinguishes an aircraft levelling off from one that will keep climbing.
Total turn over the window	Catches a long steady turn that a single snapshot would miss.
Turn rate, ignoring direction	How sharply it is turning regardless of which way.
Climb rate, ignoring direction	How steeply it is changing height regardless of up or down.
Distance to Mumbai airport	Near the airport means turning, queuing and descending. Far away means cruising.
Direction to the airport (two numbers)	Separates an aircraft heading towards Mumbai from one heading away or passing by. Stored as two numbers so that a compass direction of 359° and 1° are treated as neighbours, not opposites.
Sideways force estimate	Turn rate multiplied by speed — a measure of how hard the aircraft is actually turning.

> **WHY THESE AND NOT JUST POSITION**
>
> Our first model was given only position, height, speed and direction. None of those describes *change*. The model was never told "this aircraft is turning" — it had to work that out for itself from a sequence of compass readings, and it could not. Adding the measurements above is a large part of why the corrected model works.

SECTION 5

What the model is

The model is a type of AI called an **LSTM**. The name stands for Long Short-Term Memory, and the idea is simple: it reads a sequence one step at a time and carries a memory forward, deciding as it goes what is worth remembering and what to forget. It suits our problem because our input is a sequence — ten snapshots in order — and the order matters.

```
The 10-step movie  →  LSTM reads it front to back, building a summary
That summary + the 13 summary numbers  →  a small combining layer
→  four separate output groups, one each for 1, 3, 5 and 10 minutes
```

The model has about 300,000 internal settings, all adjusted automatically during training. Training takes a few minutes on an ordinary graphics card.

Why we removed "bidirectional"

Our proposal specified a **Bi-LSTM** — a version that reads the sequence forwards *and* backwards. The stated reason was that reading backwards lets the model "see future context".

That reasoning is mistaken. Reading backwards only re-reads the same ten *past* snapshots in reverse order. There is no future inside the input to be read — when the system runs for real, the future has not happened yet. The backwards pass therefore re-reads information the forwards pass already has.

We tested this rather than assuming it. Two models were trained, identical in every respect except direction:

BIDIRECTIONAL VERSUS ONE-DIRECTIONAL — SAME DATA, SAME SETTINGS

LOOKING AHEAD	ONE-DIRECTIONAL	BIDIRECTIONAL
1 minute	99.8%	99.8%
3 minutes	**94.1%**	93.8%
5 minutes	84.6%	84.6%
10 minutes	**58.0%**	56.7%

The bidirectional version used two and a half times as many internal settings, took twice as long to train, and was slightly *worse*. We kept the simpler one.

SECTION 6

What comes out of the model

This is the part most often misunderstood, so it is worth being precise.

THE KEY POINT

The model does not output a latitude and longitude. It outputs a *correction* to the simple straight-line guess. The full pipeline then turns that correction into a latitude and longitude.

The model produces **12 numbers** per aircraft: four time points, three numbers each.

THE THREE NUMBERS PRODUCED FOR EACH OF THE FOUR TIME POINTS

NUMBER	MEANING
Ahead / behind	How many metres further along, or short of, its straight-line path the aircraft will really be.
Left / right	How many metres to the side of that straight line it will really be. This is the number that captures a turn.
Higher / lower	How many metres above or below the simple height estimate it will really be.

Why it is built this way

Asking "where will this aircraft be?" forces the model to work out the geography and the arithmetic from scratch. Asking "how wrong will the straight-line guess be?" leaves the arithmetic to arithmetic, and leaves the model only the genuinely difficult part.

A BUILT-IN SAFETY PROPERTY

If the model has nothing useful to add, it outputs zero — and zero means "the straight-line answer is fine, use that". **The model therefore cannot perform worse than plain arithmetic.** It starts on the safe answer and only moves away from it when it has evidence.

A real worked example

An aircraft from our test data, mid-turn, at 3,360 metres and 158 metres per second. Following it through all three stages:

STAGE 1 — THE RAW 12 NUMBERS THE MODEL PRODUCES

TIME POINT	AHEAD/BEHIND	LEFT/RIGHT	HEIGHT
1 minute	−2.095	5.937	−0.403
3 minutes	−0.877	4.937	−1.774
5 minutes	−0.928	4.575	−2.251
10 minutes	−1.206	3.617	−3.055

These are on an internal scale and carry no units yet.

STAGE 2 — CONVERTED INTO METRES

TIME POINT	AHEAD/BEHIND	LEFT/RIGHT	HEIGHT
1 minute	−1,479 m	+5,271 m	−31 m
3 minutes	−4,065 m	+20,939 m	−516 m
5 minutes	−10,021 m	+38,561 m	−1,210 m
10 minutes	−34,022 m	+72,988 m	−3,810 m

Read the last row as: *"in ten minutes this aircraft will be 34 km short of where a straight line says, 73 km to the right of it, and 3,810 m lower."* The model has recognised both the turn and the descent.

STAGE 3 — ADDED TO THE STRAIGHT-LINE GUESS, GIVING REAL POSITIONS

TIME POINT	STRAIGHT LINE SAYS	OUR SYSTEM SAYS	TRUTH	SIMPLE ERROR	OUR ERROR
1 min	19.3368, 72.7853	19.3073, 72.8270	19.3008, 72.8274	3.22 NM	**0.39 NM**
3 min	19.4954, 72.8505	19.3933, 73.0222	19.3669, 73.0197	12.32 NM	**1.59 NM**
5 min	19.6540, 72.9157	19.4448, 73.2233	19.4362, 73.2225	21.77 NM	**0.52 NM**
10 min	20.0505, 73.0787	19.5284, 73.6090	19.6374, 73.7724	46.46 NM	**11.34 NM**

SECTION 7

What the complete pipeline produces

The model is one stage inside a longer chain. End to end, the system does this:

1. Collect live position reports from OpenSky for every aircraft near Mumbai
2. Group them by aircraft and keep the most recent ten reports for each
3. Work out the straight-line guess for 1, 3, 5 and 10 minutes ahead
4. Convert the ten reports into the 20 inputs described in Section 4
5. The model outputs its 12 corrections
6. Add the corrections to the straight-line guess → predicted latitude, longitude and height
7. Repeat the prediction every 15 seconds out to 10 minutes, giving a full predicted path
8. For every pair of aircraft, find the moment their two predicted paths come closest
9. If that closest approach is inside the alarm limits, raise a warning

Step 7 matters more than it looks. Earlier we only checked four moments — 1, 3, 5 and 10 minutes. Two aircraft passing dangerously close at, say, four minutes would simply not be noticed, because nobody was looking at four minutes. Checking every 15 seconds removes that blind spot.

The final output of the whole pipeline, for each pair of aircraft, is:

- the closest distance the two are predicted to come to each other,
- how many seconds from now that happens,
- the height difference at that moment,
- and whether that combination breaks the separation rules.

SECTION 8

Results

All figures below come from the 27,868 test examples — flights the model never saw during training, recorded on later days than anything it learned from.

MAIN RESULT — HOW OFTEN THE PREDICTION LANDS WITHIN 5 NM OF THE TRUTH

LOOKING AHEAD	EXAMPLES	SIMPLE METHOD	OUR MODEL	IMPROVEMENT
1 minute	27,163	99.93%	**99.96%**	—
3 minutes	23,312	91.49%	**93.51%**	+2.0
5 minutes	20,113	79.28%	**84.92%**	+5.6
10 minutes	14,030	51.82%	**64.47%**	+12.7

AVERAGE ERROR, AND ACCURACY AT A TIGHTER TOLERANCE

LOOKING AHEAD	SIMPLE, AVERAGE ERROR	OUR MODEL, AVERAGE ERROR	OUR MODEL, WITHIN 1 NM
1 minute	0.22 NM	**0.19 NM**	95.1%
3 minutes	1.40 NM	**1.13 NM**	75.9%
5 minutes	3.41 NM	**2.49 NM**	59.6%
10 minutes	11.02 NM	**6.41 NM**	25.9%

At ten minutes the average error is **almost halved**, from 11.02 to 6.41 nautical miles.

Results broken down by what the aircraft was doing

An average hides a great deal. This is where the honest picture is.

PERCENTAGE WITHIN 5 NM, BY FLIGHT SITUATION

SITUATION	EXAMPLES	1 MIN	3 MIN	5 MIN	10 MIN
Flying straight	20,071	100.0%	96.1%	88.7%	69.5%
Gentle turn	3,157	99.9%	90.7%	81.5%	58.9%
Moderate turn	1,719	99.9%	85.1%	73.0%	50.4%
Sharp turn	2,216	99.8%	79.6%	60.9%	**24.5%**
Flying level	18,020	100.0%	97.2%	92.2%	75.9%
Gentle climb or descent	5,979	99.9%	86.2%	70.6%	**41.2%**
Fast climb or descent	3,164	99.9%	85.7%	67.7%	**32.5%**
Near the airport	6,549	99.9%	83.9%	68.9%	**38.2%**
Far from the airport (cruising)	17,319	100.0%	96.3%	89.1%	70.1%

The pattern is consistent. **Up to about three minutes, everything works.** Beyond five minutes, aircraft that are turning, changing height, or close to the airport become much harder, and by ten minutes we are wrong more often than right for those.

SECTION 9

Does it actually catch conflicts?

Predicting positions is only useful if it lets us spot two aircraft heading for the same place. We tested that directly.

We took every pair of aircraft in our test data that was more than 5 NM apart and within 90 NM of each other — **273,794 pairs** — worked out which ones genuinely came dangerously close, and checked whether the system would have predicted it.

The first attempt used the exact legal minimum as the alarm trigger, and caught only 46%. That is the wrong way to set an alarm: a prediction 1 NM out flips the answer. Real air traffic systems set the alarm zone *wider* than the legal minimum for exactly this reason. Doing the same:

CONFLICTS CAUGHT, AND THE FALSE ALARMS THAT COME WITH THEM

LOOKING AHEAD	ALARM ZONE	CONFLICTS CAUGHT	FALSE ALARMS PER 1,000 PAIRS
1 minute	5 NM / 1,000 ft	46%	0.5
	8 NM / 1,500 ft	**99%**	32
3 minutes	5 NM / 1,000 ft	41%	1.2
	8 NM / 1,500 ft	**93%**	34
5 minutes	8 NM / 1,500 ft	82%	36
	12 NM / 2,000 ft	88%	50
10 minutes	8 NM / 1,500 ft	**69%**	39
	18 NM / 2,500 ft	**84%**	90

> **THE HEADLINE RESULT**
>
> With a sensibly sized alarm zone we catch **99% of conflicts one minute ahead and 93% three minutes ahead**. That meets the accuracy target set at proposal stage — provided the target is stated as *conflicts caught*, which is the thing that actually matters for safety, rather than as raw position accuracy.

The last column is the price. Widening the alarm zone catches more real conflicts and produces more false ones. That column is a dial we can set deliberately, and report honestly, rather than something we have to guess at.

One further finding worth recording: at short range, the *simple* method actually caught more conflicts than the AI (65% against 46% at one minute). At one minute the simple method is already almost perfect, and the AI's small corrections nudge both aircraft slightly, which can push a genuinely close pair apart on paper. **We therefore intend to use the simple method for the short-range alarms and the AI for the longer-range watch list.** Real systems keep a simple, predictable safety net for the same reason.

SECTION 10

The honest limit: aircraft landing at Mumbai

An aircraft arriving at Mumbai does not fly straight to the runway. The controller turns it away from the airport, runs it along a parallel leg, then turns it back to join the queue. From the outside it looks as though it is flying straight — right up to the moment it turns.

ARRIVING AIRCRAFT ONLY (WITHIN 60 NM OF MUMBAI, DESCENDING) — WITHIN 5 NM

LOOKING AHEAD	EXAMPLES	SIMPLE METHOD	OUR MODEL
1 minute	3,040	99.7%	**99.9%**
3 minutes	2,360	71.0%	**81.7%**
5 minutes	1,984	39.7%	**65.0%**
10 minutes	1,355	**1.4%**	**30.0%**

Ten minutes ahead, the simple method is right for **1.4%** of arriving aircraft. Our model reaches 30% — more than twenty times better, and still wrong about seven arrivals in ten.

Is this our model being bad, or is the future genuinely uncertain?

We measured this rather than guessing. For each arriving aircraft we found the 25 most similar situations in our historical data — same position, same height, same heading, same rate of descent — and asked whether those 25 aircraft all ended up in the same place ten minutes later.

> WHAT WE FOUND
>
> Two aircraft in an **identical** situation end up an average of **7.9 nautical miles apart**, and **64% of such pairs finish more than 5 NM from each other**.
>
> The same starting situation genuinely leads to different outcomes. This is not a modelling failure. Whether that aircraft is turned now or in two minutes is a decision a human controller makes, based on information we do not receive.

We also confirmed what drives it. In quiet periods our accuracy on arrivals is 39.8%; in busy periods it falls to 24.9%. When Mumbai is busy, controllers stretch and vector aircraft to build spacing in the landing queue, and they become far less predictable.

Why this points to a different kind of model

If several outcomes are genuinely possible, a system forced to name one will often name the wrong one. But if it offers several, the right one is usually among them:

ARRIVING AIRCRAFT, TEN MINUTES AHEAD — IS THE CORRECT ANSWER AMONG THE OPTIONS OFFERED?

NUMBER OF POSSIBLE PATHS OFFERED	CORRECT ANSWER IS AMONG THEM
1 — what we do today	**33.9%**
3	41.8%
5	50.8%
10	**64.1%**
25	**78.7%**

This matters because **for detecting conflicts we do not need to know which path the aircraft takes.** We can check all the plausible paths, and raise a warning if any of them brings two aircraft too close. Safety comes from covering the possibilities, not from guessing correctly. Section 12 makes this our main next step.

SECTION 11

What has changed from the proposal, and why

This section exists because our proposal committed us to specific choices, and several of them have changed. Each change was made because a measurement showed the original choice was wrong, and each measurement is repeatable.

EVERY DEVIATION FROM THE APPROVED PROPOSAL

ITEM	PROPOSAL SAID	WE NOW DO	WHY, AND THE EVIDENCE
Model direction	Bi-LSTM, reading the sequence forwards and backwards	One-directional LSTM	Reading the past backwards cannot reveal the future. Tested directly: bidirectional scored 56.7% at ten minutes against 58.0% one-directional, using 2.5× the settings. Section 5.
What the model predicts	Future latitude, longitude and height directly	A correction to the straight-line guess	Predicting position directly forced the model to relearn basic arithmetic and it never did. Predicting a correction also guarantees it cannot do worse than the simple method. Section 6.
How the model is scored during training	Standard squared-error scoring	Absolute-error scoring	Squared-error scoring is dominated by a small number of very wrong cases, and improved the average while collapsing the number of near-perfect predictions from 22.4% to 2.5%. Measured on our own data.
Radar noise simulation	Add artificial Indian radar noise during training	Not done during training; planned as a robustness test instead	Our data is aircraft self-reported position, not radar. We have no measurement of Indian radar noise, so any noise we invented would be an assumption we could not defend. Testing how accuracy degrades under added noise gives the same insight honestly.
Conflict warning horizon	A single conflict check out to 10 minutes	Two tiers: alarms up to 3 minutes, advisory watch list from 3 to 10	Our 10-minute accuracy does not justify telling an aircraft to change course. Real systems split the same way — see Section 12 and references 10 and 11.
Closest-approach check	Checked at 1, 3, 5 and 10 minutes	Checked every 15 seconds out to 10 minutes	Two aircraft closest at four minutes were invisible, because nothing looked at four minutes. Section 7.
How accuracy is reported	Average error only	Per time point, per flight situation, and against the simple method every run	A single averaged figure hid the fact that our first model was worse than arithmetic at every time point. That went unnoticed for weeks.

A POINT OF ACADEMIC HONESTY WE MUST STATE

We have read all five research papers cited in our proposal in full. **None of them proposes, trains or evaluates a flight-path prediction model.** Two are about converting controller–pilot radio speech into text; one is about controller staffing; two are literature summaries, and one of those is not peer reviewed.

In particular, the Bi-LSTM choice cannot be traced to any of them — the word "bidirectional" appears once across all five, in the speech paper, describing why the authors *removed* it. Our 1/3/5/10-minute choice is likewise not from any of them.

We are correcting this by citing the genuine flight-path prediction literature, listed in the references at the end of this report, and by stating plainly that our architecture is our own design justified by measurement rather than borrowed from a paper.

SECTION 12

The plan for this step

Six actions, in priority order. The first two are the important ones.

1. Split conflict detection into two tiers

Alarms and suggested course changes only from predictions up to three minutes ahead, where we are 93–99% accurate. From three to ten minutes, show the controller "watch this pair" and never generate an automatic instruction.

This is not a retreat. It is how certified systems are built. Europe's short-range safety net looks about two minutes ahead using radar only; its medium-range tool looks up to twenty minutes ahead but needs filed flight plans and is explicitly described as a planning aid rather than a safety net (references 10, 11). We have no flight plans, so the two-tier split is the correct architecture for our data.

2. Move from one prediction to several, with probabilities

Instead of one answer, the model should offer a small number of plausible paths with a likelihood attached to each — "probably continues straight, but a 30% chance it turns onto the downwind leg". Conflict checking then runs over all of them and warns if any one produces a conflict.

Our own measurement in Section 10 shows the correct answer is among ten offered options 64% of the time, against 33.9% when forced to pick one. Published work in terminal airspace reports the same: a probabilistic model reduced average prediction error by 60.3% against the equivalent single-answer model (reference 4). References 1 and 4 are the methods to follow.

3. Teach the model Mumbai's standard arrival routes

Aircraft arriving at Mumbai repeat each other's paths. We proved this with a deliberately crude test: a simple lookup of "what did the 25 most similar past aircraft do next" *beat our trained model* on arriving traffic — 33.9% against 30.0%. The proper version groups historical flights into their common routes and tells the model which route an aircraft is on. References 5, 6, 7 and 8 all use this approach successfully.

4. Add the surrounding traffic as an input

The model currently sees each aircraft alone. But whether an arriving aircraft is turned early or late depends on how many aircraft are ahead of it in the queue — we measured accuracy dropping from 39.8% in quiet periods to 24.9% in busy ones. Reference 2 does exactly this, predicting several aircraft jointly and using the airport's typical flight patterns as a guide.

5. Get more data, and get destination information

OpenSky provides free historical access to university researchers, going back several years, and its flight records include an estimated destination airport for each flight. Destination is the single most valuable missing piece of information for long-range prediction. This should be applied for now, as approval takes time.

6. Build artificial conflicts from real flights

For the next stage of the project we need examples of aircraft in conflict, and genuine near-misses are rare by design. The right method is to take two *real* Mumbai flight paths and shift one in time so that they cross. Real aircraft behaviour, artificial pairing. We should not invent artificial flight paths, because a model trained on invented turns learns our assumptions rather than Mumbai's.

> **WHAT WE EXPECT TO BE ABLE TO CLAIM AT THE END**
>
> Not "95% accurate at all time ranges" — that is not achievable and could not be defended. Instead: **95% or more of conflicts detected up to three minutes ahead, at a stated false-alarm rate and a stated warning time**, with longer-range prediction presented as an advisory tool whose limits we have measured and explained. That is a stronger result than a number we cannot support.

REFERENCES

Sources, and what each one supports

1. Barratt, S. T., Kochenderfer, M. J., & Boyd, S. P. (2019). *Learning Probabilistic Trajectory Models of Aircraft in Terminal Airspace From Position Data.* IEEE Transactions on Intelligent Transportation Systems. arXiv:1810.09568.

   Supports plan item 2 — offering several possible paths with probabilities, learned from position data alone, in exactly our setting.

2. Zhang et al. (2025). *Joint prediction of multi-aircraft trajectories in terminal airspace: A Flight Pattern-Guided Social Long-Short Term Memory network.* Engineering Applications of Artificial Intelligence.

   Supports plan items 3 and 4 — predicting several aircraft together, guided by the airport's standard flight patterns.

3. Sun et al. (2024). *Aircraft trajectory prediction in terminal airspace with intentions derived from local history.* Neurocomputing, 128843.

   Supports plan item 3 — recovering an aircraft's likely intention from historical traffic when no flight plan is available.

4. (2025). *Bayesian Spatio-Temporal Trajectory Prediction and Conflict Alerting in Terminal Area.* Aerospace, 12(9), 855.

   Supports plan item 2 — reports a 60.3% reduction in average error for a probabilistic model against the equivalent single-answer model, in a busy terminal area.

5. Gariel, M., Srivastava, A. N., & Feron, E. *Trajectory Clustering and an Application to Airspace Monitoring.* NASA DASHlink.

   Supports plan item 3 — the standard method for grouping historical flights into common routes.

6. Wang, Z. et al. (2017). *Short-term 4D Trajectory Prediction Using Machine Learning Methods.* SESAR Innovation Days.

   Supports plan item 3 — groups arrivals into routes and trains a separate model per route, at Beijing Capital airport.

7. (2025). *4D trajectory prediction and conflict detection in terminal areas based on an improved convolutional network.* PLOS ONE.

   Supports plan item 3 — route grouping followed by per-route prediction, evaluated on 8,677 arrival flights.

8. Zeng et al. *A Deep Learning Approach for Aircraft Trajectory Prediction in Terminal Airspace.*

   Background for the terminal-area prediction problem generally.

9. Mercat, J. et al. *Kinematic Single Vehicle Trajectory Prediction Baselines.* arXiv:1908.11472. And: *Physics Enhanced Residual Learning (PERL).* arXiv:2309.15284.

   Supports our Section 6 design — predicting a correction to a simple physical model rather than predicting position directly.

10. EUROCONTROL / SKYbrary. *Short Term Conflict Alert (STCA).*

    Supports plan item 1 — the short-range safety net uses about a two-minute look-ahead on radar data alone, and excessive false alarms are noted as destroying controller confidence.

11. EUROCONTROL / SKYbrary. *Medium Term Conflict Detection (MTCD).*

    Supports plan item 1 — the longer-range tool looks up to twenty minutes ahead, requires flight plans, and is explicitly "a controller tool", not a safety net.

12. Degas, A. et al. (2022). *A Survey on Artificial Intelligence (AI) and eXplainable AI in Air Traffic Management.* Applied Sciences, 12(3), 1295.

    From our original proposal. A literature survey; contributes no model or measurements of its own, but is a sound bibliography and is where references 1 and 3 were located.

13. Jamal, M., Ashraf, H., & Jhanjhi, N. Z. (2023). *Machine Learning-based Aircraft Conflict Prediction: A Systematic Literature Review.* Preprints.org, doi:10.20944/preprints202312.1600.v1.

    From our original proposal. Records that LSTM models have a small input window and are more accurate at short range than long range — consistent with our findings. Marked by the publisher as NOT PEER-REVIEWED, which should be stated whenever it is cited.

> **REPRODUCIBILITY**
>
> Every figure in this report was produced from the project's own committed code and its recorded test data. The evaluation output files are stored alongside the data, and the full evaluation can be re-run with a single command.